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Designing Applications for All Palm Platform Expansion Cards

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Learn how to help your application take better advantage of the Expansion & VFS Managers while avoiding some of the most common pitfalls.



Agenda

Quick Overview of Expansion & VFS



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- Quick Overview of Expansion & VFS
- Design Considerations



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- Technical Solutions



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- Other Developer Help



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Quick Overview of Expansion & VFS

What are Expansion & VFS?

How do you use them?

Why should you use them?



Quick Overview of Expansion & VFS

What are Expansion & VFS?

- Expansion Manager
  - Manages removable devices
  - Notifies rest of system when devices come and go
  - Provides information about these devices
- VFS Manager
  - Access to removable storage media
  - Automatic mounting and unmounting
  - Automatic launching of start applications
  - Unified API for any filesystem



Quick Overview of Expansion & VFS

How do you use them?

- Receive notifications
  - Volumes mounting & unmounting
  - Devices coming & going
- Familiar API for file access
  - Open, Close, Read, Write, Directory Enumeration
- Other APIs
  - Check for presence of devices & volumes
  - Detailed information about devices
- Low Level Access



Quick Overview of Expansion & VFS

Why should you use them?

- Many of the new Palm Powered™ Handhelds have some form of removable media
  - Palm, Sony, Handera, Acer
  - Some Springboard modules include support
  - …More on their way
- Extends the storage capabilities of the handheld for large data sets
- New distribution medium



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Design Considerations

- Cards are not Memory
- Copying vs. Moving
- Accessing Databases on a Card



Design Considerations

Cards are not Memory

- Cards are like floppy drives
  - Slow
  - Poor random access
- Cards do not replace or extend the storage heap
  - Not memory mapped
  - Slow to read, really slow to write
  - Limited bandwidth
  - Paging isn’t fun



Design Considerations

Copying vs. Moving

- Copying
  - Much like beaming
  - Copies application, bundled databases, & overlays
  - Leaves user data in storage heap on device
- Moving
  - Also brings along user data
  - Requires application intelligence for best experience



Design Considerations

Accessing Databases on a Card

- VFS Database Convenience APIs
  - VFSFileDBInfo, VFSFileDBGetRecord, etc...
    - Behavior similar to Data Manager APIs
  - For convenience only, not for heavy-duty use
    - Cards are not memory
  - Work well for querying databases
    - Ex: Getting the database names from multiple pdbs
  - Read-only access



Design Considerations

Accessing Databases on a Card

- When is it not appropriate to use prcs & pdbs?
  - If write access is required
  - Frequent random access
  - Databases that are rarely ever synced to the handheld itself
  - Databases that would be too large to be synced to most devices



Design Considerations

Accessing Databases on a Card

- Alternatives
  - Create your own file format
    - Tailor it to your applications data and access needs
  - Write your own database access functions
    - Can optimize for your particular usage scenario



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Technical Solutions

- Bundled Databases
- Launch Codes
- Notifications
- Default Directories
- Gotchas and Performance Tricks

Technical Solutions

Bundled Databases

- What are they?
  - Read-Only databases an application uses for persistent data
    - Ex: Maps, Game levels, Dictionary
- What happens with them?
  - Beamed & copied together with the application
- How to use them?
  - Set the “bundled bit” in the database attributes



Technical Solutions

Bundled Databases

- When probably not to use them:
  - Really large databases (Ex: U.S. Street Atlas)
  - Lots of databases (Ex: doc books)
  - Modifiable data (Ex: Address DB)
- What to do for these cases?
  - Distribute on a card, use VFS routines to access DB
  - Should provide a “browser” that knows about cards
  - Use CardLaunch launchcode to know you’re being run from a card, then handle getting and saving the databases yourself



Technical Solutions

Launch Codes

- sysAppLaunchCmdSyncNotify
  - Given anytime your app, or a databases for your app is copied in
- sysAppLaunchCmdCardLaunch
  - Given anytime app is run from a card
  - Given just before the normal launch
  - May only be a sub-launch
    - No UI, No Globals, Little Stack
  - Provides volRef, path, & some special flags
- sysAppLaunchCmdNormalLaunch
  - The “normal” launch code



Technical Solutions

Launch Codes

- Properly handling the “CardLaunch”
  - Start.prc applications:
    - No Globals and No UI
    - Can save info in prefs to handle during normal launch
  - Most applications:
    - Can treat as a normal launch if sysAppLaunchStartFlagAutoStart not set in flags
    - Can set SysAppLaunchCmdCardType.err = errNone before returning to prevent NormalLaunch
  - Safest approach:
    - Always assume you’re a start.prc & save info in prefs to handle later during the normal launch



Technical Solutions

Notifications

- sysNotifyVolumeMountedEvent / sysNotifyVolumeUnmountedEvent
  - Sent when a volume is mounted or unmounted
  - Applications that only deal with storage media should use these notifications
- sysNotifyCardInsertedEvent / sysNotifyCardRemovedEvent
  - Sent when a device is connected or detached
  - Drivers for IO devices & Apps providing low-level information about devices may use these notifications

Technical Solutions

Notifications

- Using them
  - Most application only need to use the volume notifications
  - Should refresh any card-related UI in response
- Issues
  - There are some types of mounted volumes that aren't associated with cards (Ex: Poser HostFS volumes)
  - Refreshing UI by calling SysUIAppSwitch on yourself doesn't always work if you are a start.prc
    - Better to simulate this behavior internally



Technical Solutions

Default Directories

- What for?
  - Some media define specific locations to place files of a specific type (Ex: jpeg files in /DCIM)
- Why?
  - Can always find files in right place, regardless of the media type
  - Better Interoperability with other devices that use the same media (Ex: Digital Cameras and MP3 players)



Technical Solutions

Default Directories

- Using them
  - Call VFSGetDefaultDirectory with the desired extension or mime type
    - Uses a volume reference number, so you don't have to know details about the actual media.
    - Returns a path to where the files should be located
  - The path provided is the "root," subdirectories may also include files of the same type

Technical Solutions

Default Directories

- Registering a new type
  - Call VFSRegisterDefaultDirectory with media type, path, and mime type or extension
    - Should only use expMediaType_any
    - Best if both mime type and extension are registered
  - Dealing with conflicts
    - Always accept the previously registered directory, or else data may appear to “disappear”

Technical Solutions

Default Directories

- Hotsync® Operation
  - Will sync all registered file types to their registered locations on a card
    - Will put an error in the log if the file type isn't registered
- Issues
  - Be careful adding new files to the locations, there may be other specifications in play
    - Ex: DCF spec for /DCIM directory
    - Hotsync doesn't take this into consideration, and always places files in the root of the registered directory



Technical Solutions

Gotchas and Performance Tricks

- Cards can be removed at any time!
  - Don’t trust that things will continue to work because they did the first time
  - Take advantage of the notifications
  - Be aware of errors returned from APIs
- Reset
  - If your app was launched from a card and it crashes, it may still be in memory after reset
    - Don’t crash ☺
    - Could delete yourself when receiving sysAppLaunchCmdSystemReset launch code



Technical Solutions

Gotchas and Performance Tricks

- Lazy Reading
  - Don’t read all the data at once in advance, only what you need to display
- Caching
  - Don’t read data more than once if you don’t have to
  - Use small “cache files” that keep important bits of information from large or multiple files handy
    - Ex: A doc reader caching the list of titles



Technical Solutions

Gotchas and Performance Tricks

- Improving launch times
  - Don’t rely on bundled databases
    - Extend your app to take advantage of the card and load these databases as needed
  - Split your data in pieces
    - Have your app intelligently load large amounts of data only as needed



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Other Developer Help

- Useful Tools
- PalmSource®
- Q & A



Other Developer Help

Useful Tools

- HostFS
  - Use with Poser
  - Maps up to 8 directories on your development machine's hard drive to simulated volumes
  - Supports dynamic mounting & unmounting
- RAMDisk
  - Simulates a real card with any device or POSE
  - Creates a record database to use for storage
  - Supports simulated insertion & removal through "custom control" calls



Other Developer Help

PalmSource

- Session 260L: Extending your Applications with Expansion Cards
  - Presented by Geoff Richmond
  - Friday @ 11:35 am
- Tonight's Open Lab



Other Developer Help

PalmSource

- What about next week?
  - Knowledge Base
  - Palm OS Developer Programs
  - We’re listening



Other Developer Help

Q & A



Other Developer Help

Q & A

- Jesse Donaldson



Other Developer Help

Q & A

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- Noah Gibbs



Other Developer Help

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- Geoff Richmond



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Q & A

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- Noah Gibbs
- Geoff Richmond
- Todd Thomas



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